Asignatura

Sistemas secuenciales programables

UNIDAD 1

Lógica digital (II)

UNIVERSAE
Instituto Superior de FP

Funciones lógicas

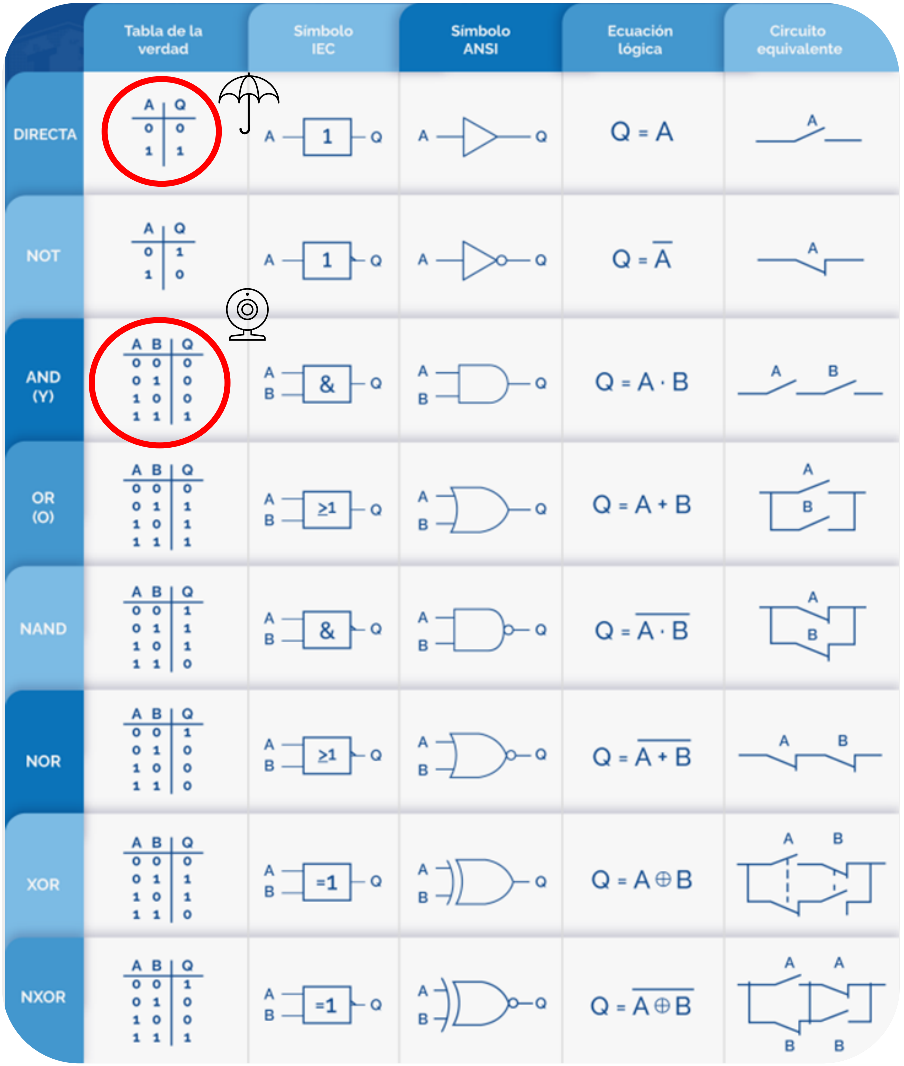
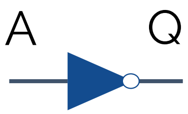


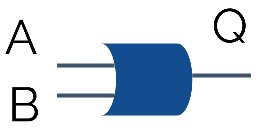
	Tabla de la verdad	Símbolo IEC	Símbolo ANSI	Ecuación lógica	Circuito equivalente
DIRECTA	A Q: 0 0; 1 1	1		$Q = A$	A
NOT	A Q: 0 1; 1 0	1		$Q = \overline{A}$	A
AND (Y)	A B Q: 0 0 0; 0 1 0; 1 0 0; 1 1 1	&		$Q = A \cdot B$	A B
OR (O)	A B Q: 0 0 0; 0 1 1; 1 0 1; 1 1 1	≥1		$Q = A + B$	A B
NAND	A B Q: 0 0 1; 0 1 1; 1 0 1; 1 1 0	&		$Q = \overline{A \cdot B}$	A B
NOR	A B Q: 0 0 1; 0 1 0; 1 0 0; 1 1 0	≥1		$Q = \overline{A + B}$	A B
XOR	A B Q: 0 0 0; 0 1 1; 1 0 1; 1 1 0	=1		$Q = A \oplus B$	A B
NXOR	A B Q: 0 0 1; 0 1 0; 1 0 0; 1 1 1	=1		$Q = \overline{A \oplus B}$	A A B B

Ejemplos circuitos lógicos

A	Q
0	1
1	0

NOT





OR

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

Ecuación a partir de tabla de la verdad

Minterms

A	B	C	F
0	0	0	**1**
0	0	1	**1**
0	1	0	**1**
0	1	1	0
1	0	0	0
1	0	1	**1**
1	1	0	0
1	1	1	**1**

$$F = \bar{A} * \bar{B} * \bar{C} + \bar{A} * \bar{B} * \mathrm{C} + \bar{A} * B * C + A * \bar{B} * C + A * B * C$$

Maxterms

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	**0**
1	0	0	**0**
1	0	1	1
1	1	0	**0**
1	1	1	1

$$F = (A + \bar{B} + C) * (\bar{A} + B + C) * (\bar{A} + B + \bar{C}) * (\bar{A} + \bar{B} + C)$$

Tabla verdad a MK (minterms)

A	B	C	D	Q
0	0	0	0	**1**
0	0	0	1	0
0	0	1	0	**1**
0	0	1	1	0
0	1	0	0	**1**
0	1	1	0	**1**
0	1	0	1	**1**
0	1	1	1	**1**
1	0	0	0	**1**
1	0	0	1	0
1	0	1	0	**1**
1	0	1	1	0
1	1	0	0	**1**
1	1	1	0	0
1	1	0	1	**1**
1	1	1	1	0

Mapa de Karnaugh

2^n

AB/CD	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	1	0	0	1
10	1	0	0	1

$\bar{A}\bar{B}\bar{C}\bar{D}$
$\bar{A}B\bar{C}\bar{D}$
$AB\bar{C}\bar{D}$
$A\bar{B}\bar{C}\bar{D}$
$\bar{A}\bar{B}C\bar{D}$
$\bar{A}BC\bar{D}$
$ABC\bar{D}$
$A\bar{B}C\bar{D}$

$\bar{D}$

$\bar{A}B\bar{C}\bar{D}$
$\bar{A}B\bar{C}D$
$\bar{A}BCD$
$\bar{A}BC\bar{D}$

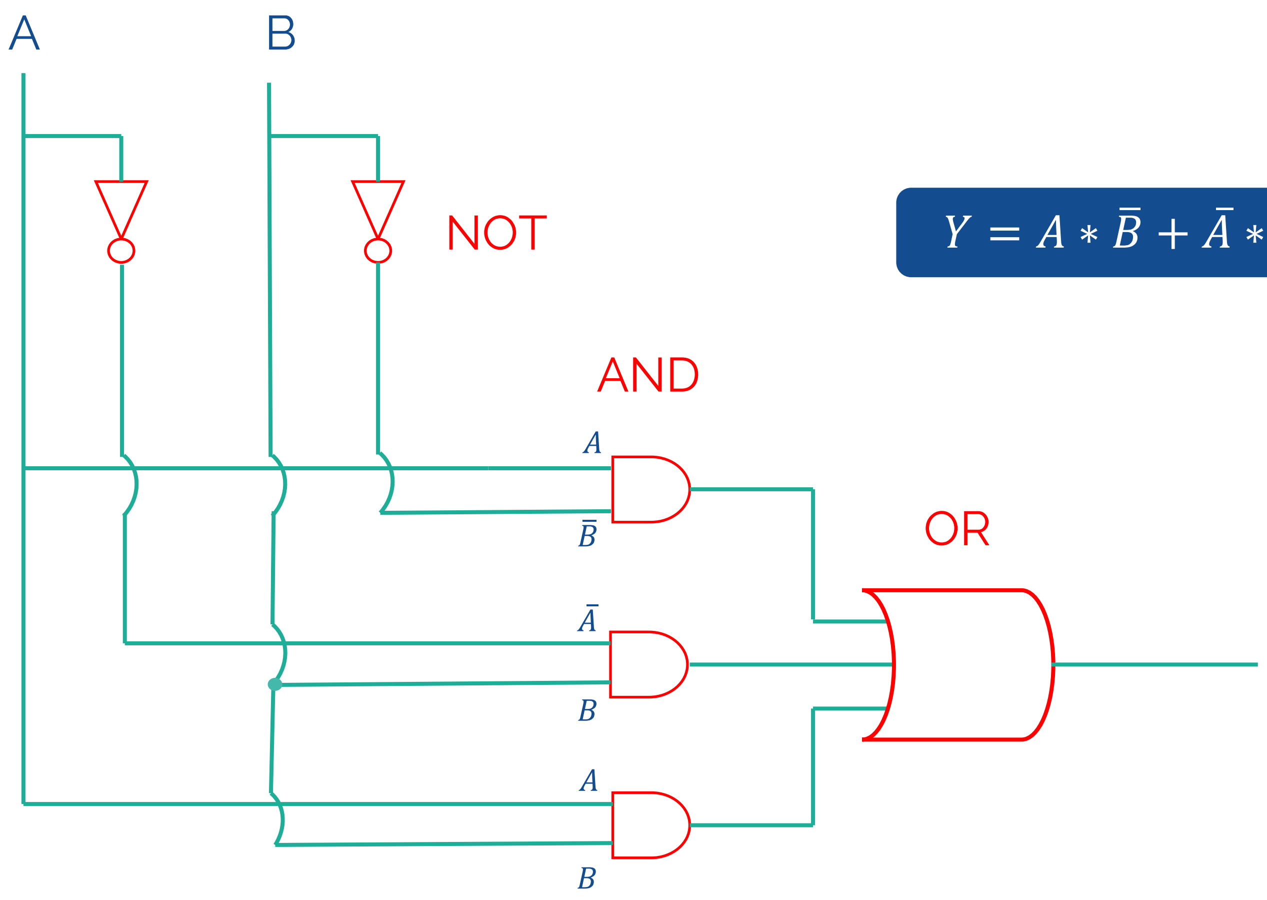
$\bar{A}B$

$$Q = \bar{A} * B + \bar{D}$$

Leyes de Morgan

$$Q = \bar{A}\bar{B}\ \bar{C}\ \bar{D}+\bar{A}\bar{B}C\bar{D}+\bar{A}B\bar{C}\bar{D}+\bar{A}BC\bar{D}+\bar{A}B\bar{C}D+\bar{A}BCD+A\bar{B}\bar{C}\bar{D}+A\bar{B}C\bar{D}+AB\bar{C}\bar{D}+AB\bar{C}D$$

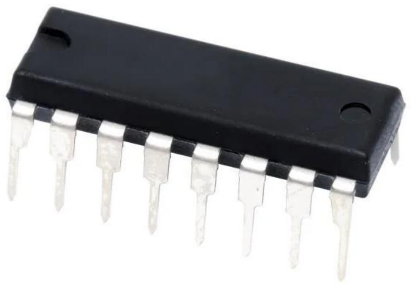
Ecuación a partir de un circuito



$$Y = A * \bar{B} + \bar{A} * B + A * B$$

Circuitos secuenciales y combinacionales

Texas Instruments



Sistemas combinacionales

Dependen solo del estado de su salida

- Multiplexores
- Demultiplexores
- Codificadores
- Decodificadores

Sistemas secuenciales

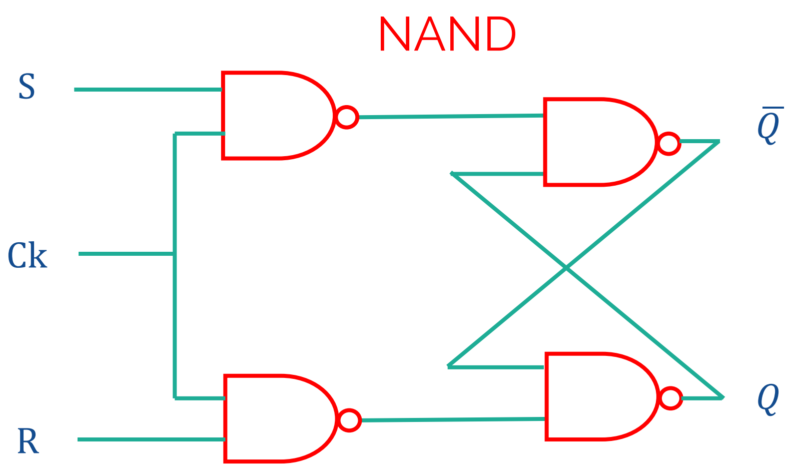
Dependen de las entradas y el estado anterior de ellas mismas

BIESTABLES

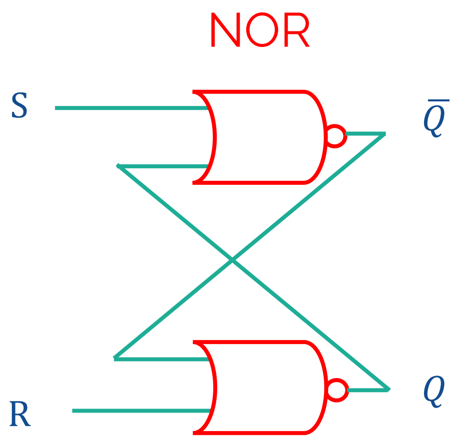
El biestable (flip-flop)

Tiene dos estados estables y almacena la información pasada (estado)

Biestable RS síncrono



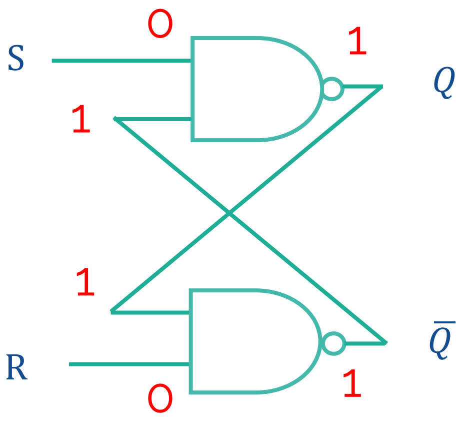
Biestable RS asíncrono

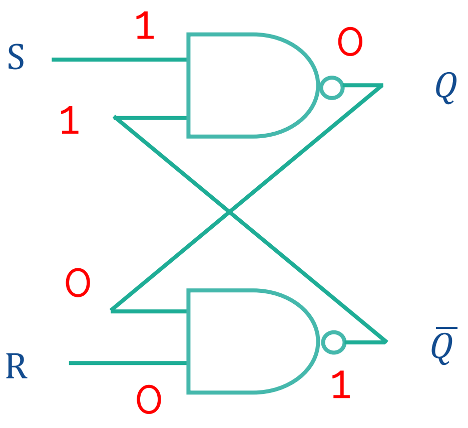


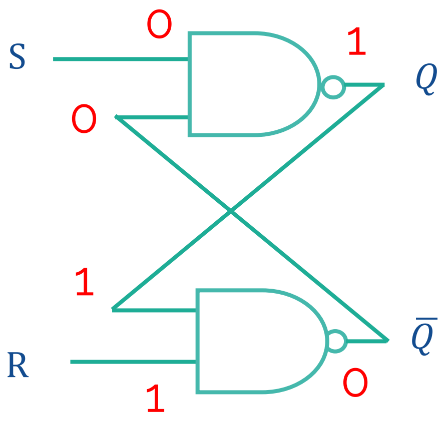
El biestable RS asíncrono

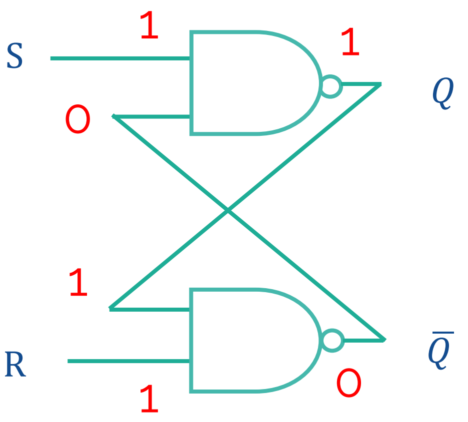
NAND

A	B	Q
0	0	1
1	0	1
0	1	1
1	1	0





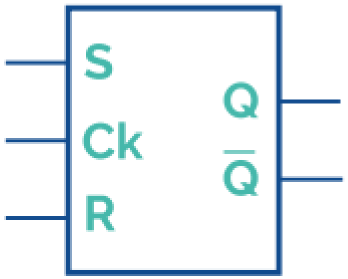




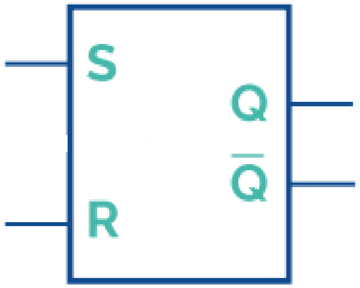
S	R	Q	$\bar{Q}$	
0	0	x	x	EP
1	0	0	1	Reset
0	1	1	0	Set
1	1	1	0	Man

La simbología de los biestables

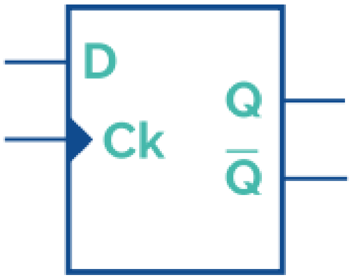
Biestable RS síncrono



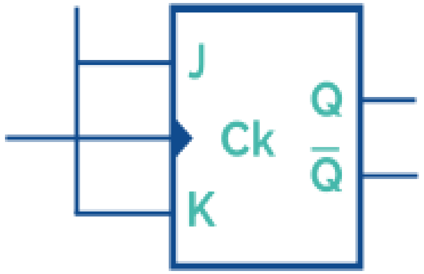
Biestable RS asíncrono



Biestable D



Biestable síncrono T



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